AIRLINE OPERATIONS & ON-TIME PERFORMANCE ANALYTICS

Portfolio Business Analytics Project • Prepared for Emirates Group Elevate Internship Application

1. Executive Summary

This project analyses airline operational performance using a 2025 flight dataset. The objective was to transform raw flight records into actionable operational insights by examining delays, cancellations, delay causes, airline-level performance and airport-level patterns. The analysis demonstrates an end-to-end workflow from data preparation through business intelligence and decision-oriented reporting.

Important: This is an independent portfolio analysis using public airline data. It does not use Emirates internal data and should not be presented as an analysis of Emirates operations.

2. Analytical Workflow & Tools

Stage	Tool / Method	Purpose
Data preparation	Python • Pandas / NumPy	Cleaned records, handled data types, removed duplicates and prepared analytical fields.
Data analysis	SQL	Structured aggregation and analytical queries for operational metrics and comparisons.
Business intelligence	Microsoft Power BI	Built interactive KPI cards, trend analysis, airline comparisons and airport analysis.
Reporting	Business analysis	Converted analytical outputs into operational findings and areas for investigation.

3. Core KPIs Tracked

Total Flights	Delayed Flights	Cancelled Flights	Flight Delay Rate	Diverted Flights
Power BI KPI	**Power BI KPI**	**Power BI KPI**	**Power BI KPI**	**Power BI KPI**

The dashboard also tracks average delay minutes per delayed flight, total delay-causing events, monthly delay patterns and the contribution of carrier, weather, NAS, security and late-aircraft causes.

4. What the Dashboard Analyses

• **Time trends:** Monthly flight-delay patterns across 2025 help identify periods requiring closer operational investigation.

• **Delay causes:** Carrier, weather, NAS, security and late-aircraft delay categories are separated to distinguish potential operational drivers.

• **Airline comparison:** Delay counts and cancellation rates are compared across carriers to identify variation that merits further investigation.

• **Airport performance:** Airport-level views examine average arrival delay, delayed-flight counts and flight volume, helping separate high-volume effects from performance patterns.

AIRLINE OPERATIONS ANALYTICS — BUSINESS FINDINGS

Interpretation of the completed Power BI dashboard • 2025 analytical scope

5. Key Findings & Business Interpretation

Finding	What the analysis shows	Business interpretation / next investigation
Delay concentration	Delay performance varies across the monthly timeline rather than remaining constant throughout 2025.	Review months with elevated delay volumes against traffic levels, weather conditions, airport congestion and schedule design.
Multiple delay drivers	The dashboard separates late-aircraft, NAS, carrier, weather and security delay contributions.	Separating causes supports targeted investigation instead of treating every delay as the same operational problem.
Airline-level variation	The dashboard compares delayed-flight volume and cancellation rate by airline.	Investigate whether differences are associated with route mix, network structure, airport exposure, scheduling or operational constraints.
Airport-level variation	Airport views compare average arrival delay, delayed flights and flight volume among high-volume airports.	Use airport-level analysis to identify locations that warrant deeper turnaround, congestion, staffing or scheduling analysis.

6. Recommended Areas for Further Analysis

• **Route-level analysis:** combine origin–destination pairs with delay and cancellation metrics to identify recurring route patterns.

• **Airport turnaround analysis:** examine taxi-out, taxi-in and late-aircraft effects to understand potential turnaround bottlenecks.

• **Operational segmentation:** compare delay performance by time of day, day of week, aircraft type and flight distance where available.

• **External factors:** enrich the dataset with weather, airport congestion and calendar/event data to distinguish operational from external drivers.

• **Customer impact:** connect operational delay metrics with passenger-volume or satisfaction data where available to estimate business/customer impact.

7. Why This Project Demonstrates Analytics Capability

The project demonstrates the ability to take a large operational dataset, structure and clean it, query it for meaningful metrics, build an interactive dashboard, and translate visual patterns into questions that an operations team could investigate. The emphasis is not only on producing charts, but on connecting data to a business decision-making process.

8. Interview Project Story

Problem: Airline delays and cancellations can have multiple operational causes and may vary by time, airline and airport.
Approach: I cleaned and prepared the data in Python, used SQL for structured analysis, and built a Power BI dashboard to monitor KPIs and identify patterns.
Analysis: I compared monthly delays, delay causes, airline performance and airport-level metrics.
Outcome: The dashboard turns raw operational records into a structured view of where performance varies and where deeper investigation could be focused.

Portfolio note: This is an independent demonstration of data analytics and business problem-solving using public airline data. Conclusions are limited to the scope and quality of the underlying dataset.